Evaluating Dataset Quality and Machine Learning Readiness with Tool-Grounded Agents

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***Abstract*—Detecting a data-quality defect does not establish that repairing it improves a machine-learning task. We evaluate a framework that connects tool-grounded assessment, constrained repair, held-out predictive utility, and reference-relative dataset readiness. Deterministic code supplies inspection evidence, validates proposed edits, and protects unauthorized values; rules, a single tool-using language model, and a specialist agent team provide alternative assessments. Existing experiments cover Adult, Diabetes, and Steel classification tasks. In a complete 27-draw missingness cohort using Logistic Regression, cell detection F1 is 1.000 throughout, yet median/mode repair reduces macro-F1 in all nine Adult draws, improves it in all nine Diabetes draws, and reduces it in seven of nine Steel draws. Mean changes are respectively -0.485, $+0.315$, and -0.108 percentage points. Native Diabetes controls further expose disagreement between macro-F1 and ranking metrics after repair. A prospectively specified six-case readiness study compares the assessors on identical observations: rules and the single model each produce four correct decisions per six assignments, while the agent team produces three and completes only three. The evidence supports evaluating detection, edit integrity, utility, and operational completion separately; the small agent study does not establish architectural superiority.**

***Index Terms*—data quality, data mining, data cleaning, dataset readiness, tool-using agents, downstream evaluation**

I. Introduction

Data validation provides executable tests for completeness, uniqueness, types, and declared ranges [1]. These tests answer whether a table satisfies a requirement. A predictive mining pipeline also needs to know whether a detected violation matters for its task, whether an available repair is appropriate, and whether making that repair changes held-out performance. A missing-value detector can be exact while imputation changes useful distributions. Conversely, a table can satisfy routine structural checks while containing a target proxy that invalidates its evaluation.

Prior research already establishes the need to evaluate cleaning through downstream learning. CleanML studies cleaning choices across classification tasks [2], while probabilistic repair systems such as HoloClean address the inference of plausible corrected values [3]. The remaining question examined here is how to connect inspection, a permitted intervention, verified execution, and a readiness decision when a tool-using language model participates in the assessment. Agent reasoning can select findings and actions, but it does not replace measured evidence. Tool-mediated interaction, exemplified by ReAct [4], makes that distinction implementable; its usefulness for dataset readiness still requires direct evaluation.

We study a pipeline in which Python owns profiling, coordinate validation, repair execution, and verification. The assessor can be deterministic rules, one tool-using language model, or three independent specialists followed by a coordinator. The evaluator retains corruption truth and downstream scores outside assessor access. Dataset readiness is defined relative to a fixed reference, task, and repair contract, with three outcomes: usable as observed, remediable, or not ready under the allowed operations. This definition makes the decision reproducible without equating it with deployment certification.

The paper makes three contributions. First, it specifies an evaluation framework linking defect localization, permitted edits, signed predictive effects, and readiness, while separating agent-visible evidence from evaluator truth. Second, it analyses an existing cross-dataset missingness cohort and declared cross-defect examples, showing that accurate detection and valid execution can accompany positive, neutral, or negative utility. Third, it evaluates the operational behavior of rules, a single model, and a specialist team on a matched readiness case study, retaining failed assignments and their recorded resource usage. The contribution is the joint evaluation of these outcomes in one guarded pipeline; heterogeneous cleaning effects themselves are established in prior work.

Four questions organize the evidence. *RQ1:* Does accurate detection guarantee beneficial repair? *RQ2:* What do verified repairs and native controls reveal about task utility? *RQ3:* Can the assessors distinguish `READY`, `REMEDIATE`, and `NOT_READY` under a fixed contract? *RQ4:* How do completion and resource usage affect the observed value of single-model and multi-agent assessment? These questions retain the original study's detection, remediation, utility, readiness, and agentic objectives. Architectural role effects and provider generalization remain outside the measured claims.

II. Related Work

A. Validation and repair

Schelter et al. [1] express data-quality requirements as constraints over computed metrics and execute validation at scale. This is the relevant foundation for our observation-only rules and deterministic inspection tools. We evaluate a bounded set of implemented checks; we do not reproduce the cited

system's complete capabilities or its scalability evaluation. HoloClean [3] combines constraints, external information, and statistical evidence in probabilistic repair. Our interventions instead use an explicit, restricted operation contract. Constraint resolution, restoration of an original value, and predictive benefit are consequently measured as distinct properties.

B. Cleaning for predictive tasks

CleanML [2] systematically examines how data cleaning affects classification, including the dependence on dataset, error type, learner, and cleaning method. We extend that evaluation perspective to readiness decisions, protected edits, and failure-inclusive agent assessment. Conformal Data Cleaning [5] also evaluates predictive consequences, with reported experiments that train on clean data and clean corrupted test inputs. Our principal interventions corrupt and repair training inputs while preserving a designated test set, so the evaluated intervention differs. Confident Learning [6] estimates label-quality information from model probabilities and is a relevant specialist approach to label errors. It is contextual literature, not an evaluated baseline here.

C. Tool-grounded assessment

ReAct [4] combines language-model reasoning with actions that obtain external evidence. Our agents similarly inspect bounded tool outputs and produce structured findings. The distinctive evaluation issue is whether those findings support allowed edits and reference-relative readiness, including execution failures. A specialist architecture is an implementation choice, not evidence of beneficial specialization. Accordingly, the operational comparison reports the actual request allocation, completion, and elapsed time rather than attributing differences causally to agent roles. None of the cited cleaning or agent systems is presented as a measured competitor.

III. Assessment and Evaluation Framework

A. Observations and information access

Let D_0 be a task-specific reference training table, D_o an observed training table produced by a recorded intervention, and T the fixed held-out table. Assessment uses D_o, its declared schema, task metadata, and public numeric bounds. Corruption manifests, original values, generating seeds, T, and downstream scores are evaluator-only. Opaque identifiers reduce explicit clues in case names. Semantic column names remain visible and can reveal benchmark identity.

Figure 1 shows the implemented path. Python supplies column profiles, bounded row inspection, missing-row localization, exact duplicate groups, numeric parseability checks, declared-range checks, and batched missing-column localization. The language model receives profiles and bounded tool responses, without arbitrary-code execution or unrestricted filesystem access. Findings identify an issue type, level, column, and host-validated coordinate set.

Rules report supported observations directly. The single-model assessor can query the inspection catalog and propose structured findings. The multi-agent assessor runs quality, schema, and metadata/ML specialists independently, followed by a readiness coordinator. Specialists do not see each other's reports. The coordinator can retain, dismiss, or flag existing candidates for review, but cannot inspect the data itself. A required-role failure fails the assessment rather than silently substituting a rules result.

The final matched case study uses Claude Sonnet 4.5 through a pinned AWS Bedrock US inference profile at temperature 0.0. A single assessment permits at most four model requests. The team permits at most ten: three per specialist and one coordinator request. Both have at most 24 tool calls. A configured pre-request reservation guard can also prevent a subsequent request from starting. Specialist tool quotas are 14, 6, and 4. No outcome-driven retries are added. These limits produce a same-input operational comparison, not a compute-matched architectural experiment.

B. Constrained intervention and verification

An assessment finding does not grant arbitrary editing permission. The executor validates the operation type, columns, and coordinate set against the current repair contract, rejects unsupported requests, and verifies protected values after execution. Targets, row identities/order, schema, and unauthorized predictor cells are protected in the readiness study. Earlier label-noise experiments separately authorize training-label changes.

Historical policies are kept separate. The cross-dataset missingness cohort uses task-specific median/mode repair. The declared-domain pilot permits masking supported negative integers in eight nonnegative Diabetes predictors and preserves natural missingness. The later readiness study masks approved negative values, then fills approved missing cells and newly masked cells using nonnegative observed training medians or categorical modes. Empty-column fallbacks are zero and a declared missing-category token. Unapproved negatives are excluded from numeric donors. Masking removes a violation; it does not recover its hidden original value.

The evaluator applies a separate canonical repair using the complete eligible observed findings under the same operation contract. This determines whether an allowed repair exists and satisfies the readiness criteria. An agent's own repair uses only its selected findings. Canonical utility is not credited to an agent that failed or omitted the action. The evaluator labels a case independently of the agent's prediction.

C. Detection and predictive effects

For an explicitly declared defect unit, detection precision and recall compare flagged units with constructed evaluator truth; detection F1 is their harmonic mean. Cells, duplicate groups, and columns represent different endpoints and are not pooled. Natural missingness is distinguished from injected missing cells where both occur. Detection evaluates localization, whereas verification evaluates allowed execution and protection of other values.

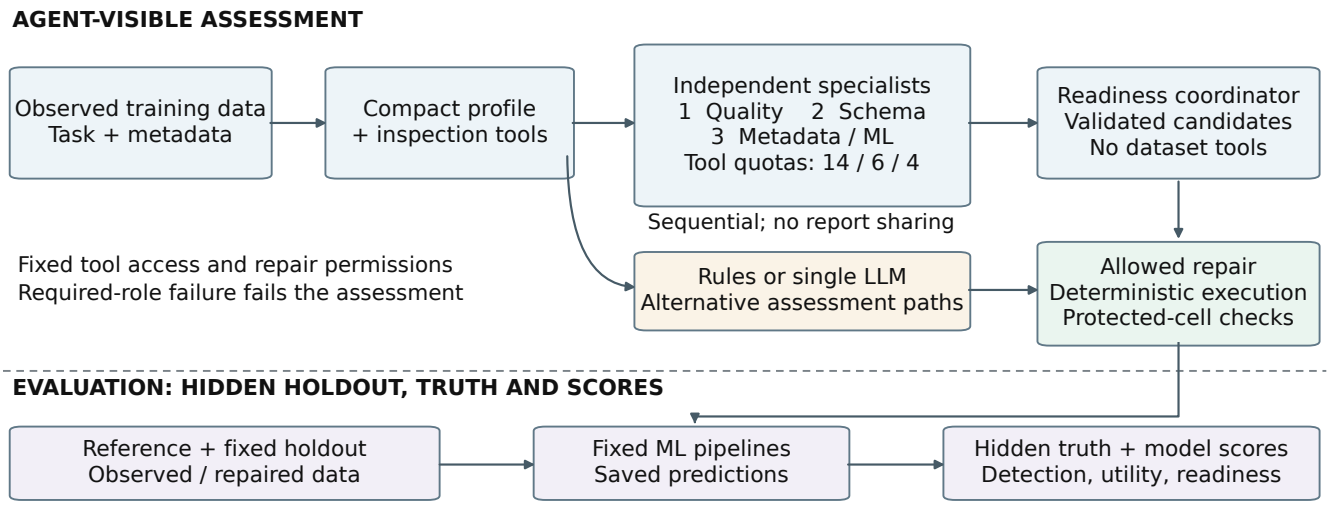


Fig. 1. Implemented assessment and evaluation boundaries. Quality, schema, and metadata/ML specialists run sequentially with independent contexts. The coordinator receives validated candidates and reports, with no inspection tools. Rules and a single tool-using LLM are alternative assessors. Python executes and verifies allowed repairs; corruption truth and held-out scores remain evaluator-only.

For a higher-is-better task metric S, define reference, observed, and repaired scores as S_0, S_o, and S_r. We retain signed degradation L, repair benefit B, and residual loss E:

$$L = S_0 - S_o, \qquad B = S_r - S_o, \qquad E = S_0 - S_r. \quad (1)$$

Positive B denotes improvement, negative B denotes harm, and zero denotes no score change. For lower-is-better log loss, directions reverse so that positive benefit still indicates improvement. Recovery is

$$\rho = B/L \quad \text{only if } L > 10^{-12} \text{ and utility is valid.} \quad (2)$$

Negative benefits and recovery above one are retained. Non-positive degradation does not have a meaningful recovered fraction. Exact target-proxy diagnostics have invalid ordinary utility and are excluded from recovery claims. Reported percentage-point effects equal $100B$ for scores on a zero-to-one scale.

D. Reference-relative readiness

Readiness uses macro-F1, ROC-AUC, and average precision (AP), with Logistic Regression as the primary label pipeline. The tolerance is 0.02 absolute score units and the severe-loss threshold is 0.10, equivalent to two and ten percentage points. Critical conditions are an unusable target, a declared numeric-domain violation, or complete predictor unavailability. Ordinary native categorical missingness alone is not critical. The evaluator applies these ordered rules:

1) Assign `NOT_READY` for an unusable supervised target or verified data-attributable training impossibility. Missing/invalid target values or absence of a required target class are unusable; targets cannot be repaired in this study.
2) Otherwise assign `READY` when every primary signed loss is at most 0.02 and no critical condition remains.
3) Otherwise assign `REMEDIATE` if the canonical repair is permitted, verified safe, clears critical conditions, and leaves every primary residual loss at most 0.02. Each metric with loss above 0.02 must also recover at least 50% of that loss.
4) If a required metric or repair-verification outcome is unavailable, abstain. Otherwise assign `NOT_READY` when any unresolved primary loss is at least 0.10; remaining cases are indeterminate.

Execution failures are distinct from `NOT_READY` truth. An `ABSTAIN` evaluator outcome would remain unverified rather than become a fabricated readiness class. All six evaluated cases have determinate truth.

The observation-only rules assessor predicts `NOT_READY` for unusable targets or all predictors missing, `REMEDIATE` for any predictor missingness or declared negative value, and `READY` otherwise. It never uses downstream scores. Thus it can disagree with task-relative truth, particularly when native missingness is tolerated. Agents also predict before receiving held-out utility.

IV. Experimental Design

A. Datasets and fixed partitions

Table I describes the processed references actually evaluated. Adult [7] predicts recorded income class from 14 predictors. Pinned source partitions are combined, representations normalized, incomplete rows and exact duplicate rows removed, and identical predictor groups kept within one side of a stratified split. The resulting complete-case population differs from the original source population.

Diabetes [8] predicts recorded readmission within 30 days from 47 predictors available at discharge. Later and no recorded readmission form the negative class. Expired/hospice

TABLE I
EVALUATED LOCAL REFERENCES. NUM/CAT COUNTS EXCLUDE THE TARGET.

Task	Train	Test	Num/cat
Adult income	36,140	9,035	6/8
Diabetes readmission	79,473	19,870	8/39
Steel Load_Type	25,662	8,648	6/2

exclusions produce 99,343 encounters; splitting is patient-disjoint and stratified. The reference preserves 188,261 naturally missing predictor cells, including 150,775 in training. This is retrospective discharge-time classification, with incomplete follow-up information, not admission-time clinical validation.

Steel [9] predicts recorded `Load_Type`. Five sensor inputs are lagged by one 15-minute interval within the same day; current time-of-day and two calendar features complete the eight predictors. Current sensor values, CO2, prior labels, and cross-day lags are excluded. January–September 2018 provides training data and October–December provides testing. Earlier observed test-period sensors can supply rolling lagged covariates without retraining. This is recorded-label classification; source-label construction and measurement latency do not support a validated forecasting claim.

B. Downstream learners

Logistic Regression (LR), Random Forest (RF) [10], and XGBoost (XGB) [11] use fixed task-specific preprocessing and configurations, with seed 42. LR uses $C = 1$, `lbfgs`, and 2,000 maximum iterations; RF uses 100 trees, depth 12, and square-root feature sampling; XGB uses 100 rounds, depth 6, and learning rate 0.1. Training-fitted preprocessing uses numeric medians, a constant categorical missing token, and one-hot encoding. Only LR standardizes numeric features. Diabetes limits each encoded categorical feature to 32 categories. No holdout tuning or early stopping is performed.

Each available reference, observed, and repaired arm is evaluated on the same task's test rows. Preprocessing changes private model matrices, not the saved assessment table. LR is primary for the missingness cohort and readiness truth; RF and XGB provide native-control and readiness sensitivity analyses. Multiple learners on one input are not independent datasets. Steel ROC-AUC and AP average one-versus-rest class scores.

C. Evidence cohorts and analysis scope

The principal missingness cohort is a complete saved slice: three datasets, rates of 5%, 10%, and 20%, and corruption seeds 42–44, giving 27 LR comparisons. Rates refer to eligible originally observed predictor cells. We use all draws in this slice, align cell detection by comparison identifier, and compute paired benefit from saved observed/repaired scores. The analysis is a retrospective synthesis of existing measurements, not a new prospective experiment. Per-severity mean and sample SD describe three perturbation draws on a reused holdout. Positive/negative counts describe finite score changes, not population probabilities or a practical-significance threshold.

To examine other mechanisms, we retain the declared Adult rules examples from the original synthesis: LR, seed 42, 10% for rate-based defects, two added features, and one exact target proxy. The eight examples include the Adult 10% missingness draw already in the 27-comparison cohort; they are not eight additional independent observations. Their selection occurred after outcomes were known. We analyse the native Diabetes control separately under the final combined repair policy and all three learners.

The matched readiness study specifies six Diabetes conditions before generating its new corruption draws: native C01; 5% predictor missingness C02; exact negation of 10% and 20% of eligible positive numeric cells C03/C04; all predictors missing C05; and all targets missing C06. Seeds are 2001 for C02/C03 and 2002 for C04. Full masks are fixed interventions, not random replications. The patient-disjoint holdout had already been used during development. All assessors receive the same observations and operation contract; only the assessor and its compute allocation differ.

For context, the historical inventory spans 194 observed tables and 684 assigned model/method comparisons: 645 completed, 36 blocked, and three unavailable. Three completed comparisons are target-proxy diagnostics with invalid ordinary utility. Its 1,349 raw records count arms and attempts, not independent experiments. These historical cohorts have different repair permissions and are not pooled into an agent ranking or one utility estimate. The supplied compact sources preserve the primary cohort summaries and readiness ledger, but not the full historical utility export or all saved predictions.

D. Operational outcomes and uncertainty

The primary readiness endpoint is verified correct decisions divided by all assigned cases. Completion is reported separately, and failed outputs remain in the denominator. Completed-only three-label macro-F1 is supplementary; a missing completed class contributes zero under the fixed-label convention. We retain model requests, tool calls, and elapsed assessment time, including incomplete attempts.

Earlier uncertainty reconstruction used 2,000 paired resamples of saved predictions, seed 450048, and pointwise 95% percentile intervals for macro-F1 effects. Adult used an IID-row approximation; Diabetes sampled patient clusters; Steel used seven-day moving blocks, following the dependence-aware resampling motivation of Künsch [12]. The archive reports 492 eligible comparisons and 192 skips. These intervals condition on fixed fitted models and reused holdouts, omitting training and selection uncertainty and multiplicity adjustment. We do not recompute intervals from aggregate scores or treat the three-seed SDs in the present analysis as confidence intervals. No new model fit or LLM call is required for this synthesis.

TABLE II
PAIRED LR MACRO-F1 BENEFIT (PP) IN THE COMPLETE MISSINGNESS SLICE. EACH CELL IS MEAN ± SAMPLE SD OVER THREE DRAWS.

Task	5%	10%	20%
Adult	-0.425 ± 0.059	-0.398 ± 0.202	-0.633 ± 0.313
Diabetes	$+0.197 \pm 0.143$	$+0.370 \pm 0.288$	$+0.379 \pm 0.166$
Steel	-0.168 ± 0.166	-0.109 ± 0.022	-0.047 ± 0.174

V. RESULTS

A. RQ1 Accurate detection does not guarantee benefit

Cell detection F1 is 1.000 in all 27 missingness comparisons, including the injected-cell scope in Diabetes, where native missingness is tracked separately. Nevertheless, the repair benefit changes sign across tasks (Fig. 2). All nine Adult draws have negative benefit, all nine Diabetes draws have positive benefit, and seven of nine Steel draws have negative benefit. Across the nine draws per task, mean effects are -0.485 pp, $+0.315$ pp, and -0.108 pp, respectively. These means summarize the declared rates equally; they do not estimate the prevalence of harmful cleaning in a target population.

Table II shows the severity-specific results. Adult's negative effect persists across every draw and becomes more negative on average at 20%. Diabetes benefits at all three rates. Steel has small positive effects in one 5% and one 20% draw, while all 10% draws are negative. The largest individual absolute effect is an Adult decrease of 0.908 pp. These are finite observed changes; repeated holdout use and only three seeds preclude a broad significance or generalization claim.

The detector identifies missing locations, but it does not determine whether the corresponding edit improves a fitted predictor. Because detection F1 is constant in this cohort, a detection–utility correlation is not estimable here. The finding is a direct counterexample to the implication that perfect detection guarantees beneficial remediation. It does not establish a calibrated predictor of cleaning value or the general independence of detection and utility.

B. RQ2 Repairs have distinct structural and predictive outcomes

Table III joins detection and utility for the declared Adult examples. A selected missingness draw moves from observed macro-F1 0.792056 to repaired 0.787876 despite perfect detection. The reference is 0.788815, so the observed corruption has negative degradation and recovery is undefined. Category canonicalization also detects all constructed variants but reduces macro-F1 by 0.037 pp; type-token repair has zero macro-F1 benefit.

Exact duplicate removal restores the reference score, giving recovery 1.000. Domain repair produces benefit 0.294 pp against degradation 0.180 pp, yielding recovery 1.638; the ratio is not clipped. Conversely, label-flip detection has precision 0.976 but recall 0.034 and F1 0.066, yet the authorized repair produces benefit 0.504 pp and recovery 0.729 in this selected comparison. The added-feature rule finds the constant column, misses the varying irrelevant column, and has no macro-F1 benefit. The proxy diagnostic obtains observed F1 1.000 through exact target leakage; that score is invalid evidence of useful generalization.

TABLE III
DECLARED ADULT RULES EXAMPLES WITH LR AND SEED 42. RATE-BASED DEFECTS USE 10%; ADDED FEATURES USE TWO COLUMNS. DETECTION UNITS ARE SEPARATE. RECOVERY REQUIRES POSITIVE DEGRADATION AND VALID UTILITY.

Defect	Unit	Det. F1	B (pp)	Recovery
Missingness	Cell	1.000	-0.418	Undef.
Duplicates	Group	1.000	$+0.140$	1.000
Type tokens	Cell	1.000	0.000	Undef.
Numeric domains	Cell	1.000	$+0.294$	1.638
Category variants	Cell	1.000	-0.037	Undef.
Label flips	Cell	0.066	$+0.504$	0.729
Added features	Column	0.667	0.000	0.000
Target proxy	Column	1.000	Invalid	Undef.

TABLE IV
NATIVE DIABETES REPAIR BENEFIT IN PERCENTAGE POINTS. THE TABLE IS ONE CONTROL INPUT EVALUATED WITH THREE FIXED LEARNERS.

Learner	Macro-F1	AP	ROC-AUC
LR	$+0.247$	-0.282	-0.200
RF	0.000	-0.373	-0.117
XGB	-0.261	-0.790	-0.307

These cases separate localization quality from task benefit without ranking incomparable defect units. They are mechanism illustrations selected after outcomes, not confirmatory comparisons between cleaners. The archived pointwise benefit interval for the domain example crosses zero (-0.177 to $+0.791$ pp), reinforcing that its positive point estimate alone does not establish reliable gain.

Native controls reveal a second separation: utility metrics can disagree even without synthetic corruption. Figure 3 and Table IV show the complete native Diabetes control under the final combined policy. LR macro-F1 increases by 0.247 pp, while AP decreases by 0.282 pp and ROC-AUC by 0.200 pp. RF macro-F1 is unchanged, but both ranking metrics decrease. XGB macro-F1, AP, and ROC-AUC decrease by 0.261, 0.790, and 0.307 pp, respectively.

Native here means no injected defect, not absence of naturally missing values. Therefore the result shows the consequence of modifying the reference's missingness representation, rather than proof that every clean dataset is harmed by cleaning. The same XGB native result is reused by rules and the completed single-model repair and is counted once. Protection checks establish which values changed; these score differences establish a separate, task-dependent consequence.

C. RQ3 Readiness depends on task loss and the operation contract

The six conditions yield two cases in each readiness class: C01/C02 are `READY`, C03/C04 `REMEDIATE`, and C05/C06

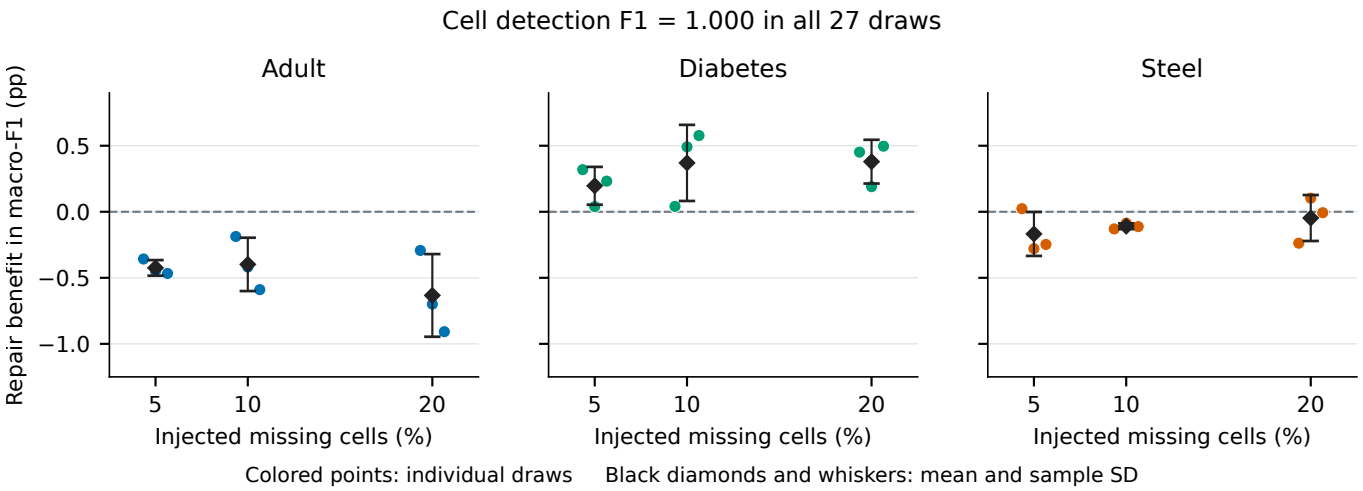


Fig. 2. All 27 saved LR missingness draws under task-specific median/mode repair. Benefits are repaired minus observed macro-F1 in percentage points (pp); all use the task's fixed holdout. Cell detection F1 is 1.000 throughout. Individual draws and mean ± sample SD are shown for each rate; SD is perturbation variation, not a confidence interval. Shared scales show that the effects are modest and differ in sign across tasks.

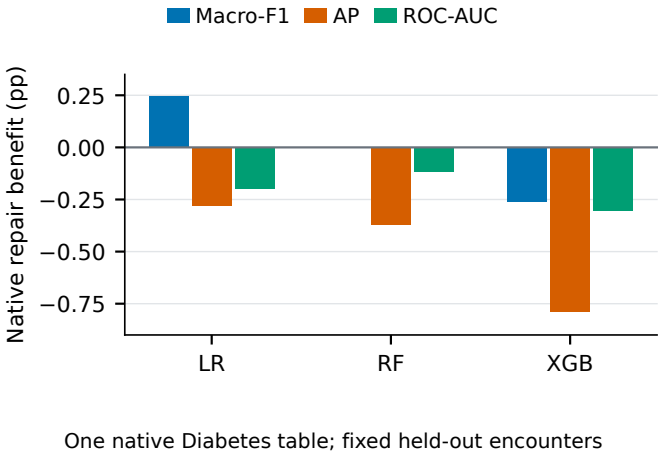


Fig. 3. Native Diabetes control under the combined repair policy. LR, RF, and XGB denote Logistic Regression, Random Forest, and XGBoost. Bars show paired score changes in pp on the same held-out encounters. A favorable macro-F1 change can accompany worse AP and ROC-AUC. Learners are sensitivity analyses on one shared input.

NOT_READY. RF and XGB sensitivity pipelines agree with LR on all six labels. They remain six cases, not 18 independent observations. Figure 4 presents every primary label and assessor outcome, including failures.

The maximum primary losses in C03/C04 are 0.437 and 0.733 pp, below the two-point tolerance. They require remediation because declared domain violations remain and the allowed canonical repair meets the residual-loss requirements. C02 remains ready despite injected missingness because its losses are within tolerance and ordinary missingness is not critical. Thus the policy does not declare every visibly imperfect table unready.

C05 illustrates why macro-F1 alone is insufficient. Removing all predictors reduces LR macro-F1 by 1.484 pp, below the two-point tolerance, but reduces ROC-AUC by 17.032 pp. No acceptable canonical repair restores the destroyed information, so the case is NOT_READY. C06 is also NOT_READY because the protected supervised target is unavailable; a provider error would not justify that label.

Rules and the single model each classify four of six assigned cases correctly; the team classifies three. Neither rules nor either live method correctly identifies a READY case. Rules over-prescribe repair on native and mildly corrupted inputs. The single model labels C01 REMEDIATE and fails on C02; the team fails on both. Both live methods correctly label the two domain cases, while the team completes only one of the two NOT_READY cases. The evidence therefore identifies a specific operational weakness in leaving usable observations unchanged, alongside successful completed decisions on repairable domain violations.

D. RQ4 Completion and resource usage change the interpretation

Table V reports the same six assignments for every assessor. The single model completes 5/6; the team completes 3/6. Completed-only three-label macro-F1 is 0.600 for the single model and 0.667 for the team, but verified correct per assigned case is 4/6 and 3/6. The latter is the primary endpoint. The team's fixed-label macro-F1 includes a zero contribution for READY, absent from its completed subset. Its three correct completed decisions do not establish better operational accuracy.

Across 12 live assignments, the assessors use 51 model requests and 132 tool calls. The single model accounts for 14 requests and 48 tool calls; the team accounts for 37 requests and 84 tool calls. Mean recorded assessment times across all attempts are 70.3 and 209.9 seconds, respectively. Three attempts terminate at output-token limits and one fails at the configured pre-request reservation guard. All four failures remain in the assignment denominator and resource summaries.

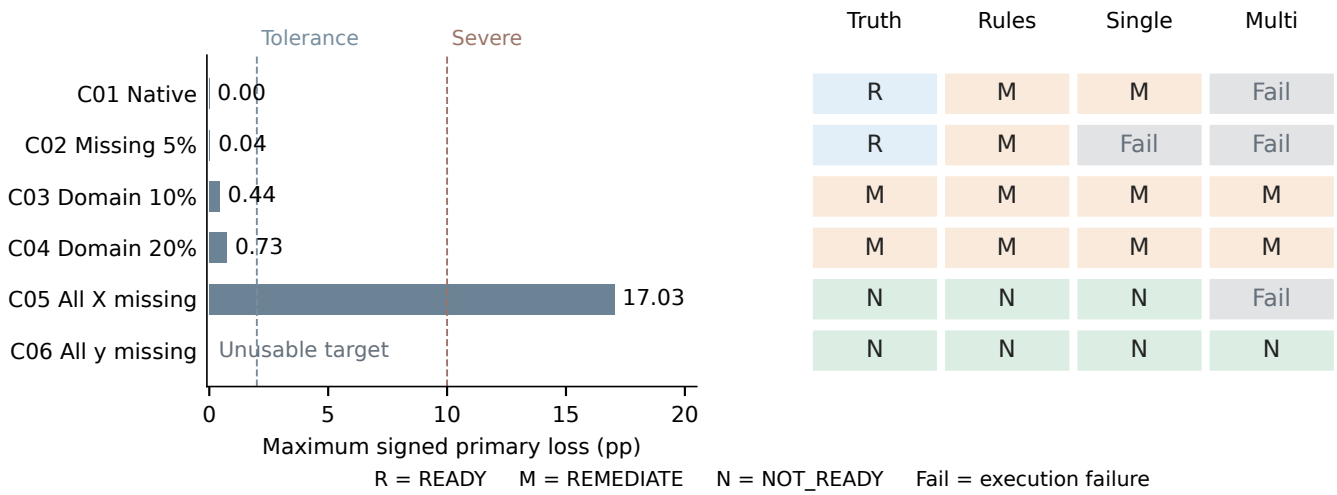


Fig. 4. Matched six-case Diabetes readiness study. Left: maximum signed LR loss across macro-F1, AP, and ROC-AUC; dashed lines mark the two- and ten-point thresholds. C06 has no valid supervised score and is displayed structurally. Right: evaluator truth and assessor predictions. Domain cases require remediation despite small losses; execution failure remains separate from dataset truth.

TABLE V
FAILURE-INCLUSIVE READINESS AND RESOURCE USAGE. CORRECTNESS USES ALL SIX ASSIGNED CASES. COMPLETED-ONLY MACRO-F1 IS SUPPLEMENTARY. MODEL REQUESTS, TOOL CALLS, AND MEAN ELAPSED TIME INCLUDE INCOMPLETE ATTEMPTS. DASHES INDICATE UNRECORDED RESOURCE MEASUREMENTS FOR DETERMINISTIC BASELINES.

Assessor	Completed	Correct/assigned	F1 completed	Model requests	Tool calls	Mean time (s)
Always `READY`	6/6	2/6	0.167	—	—	—
Rules	6/6	4/6	0.556	—	—	—
Single LLM	5/6	4/6	0.600	14	48	70.3
Multi-agent	3/6	3/6	0.667	37	84	209.9

The team uses more model requests and tool calls and takes longer on average while completing fewer assignments. These resource measurements describe this implementation and small cohort. Unequal request ceilings, different prompts, and the team's required-role failure rule prevent a causal efficiency or specialization claim. The observed team is less operationally successful here, but the study does not rank all multi-agent architectures.

The earlier domain-only feasibility pilot completes just one of six live assignments, the native single-model case. Its five failures remain part of the historical record; the later combined policy and engineering changes do not convert them into successful corrupted-domain repairs. Keeping those cohorts separate prevents a revised narrative from disguising development failures as a stable matched benchmark.

VI. DISCUSSION AND LIMITATIONS

A. Implications for data mining evaluation

The experiments motivate a decision sequence in which the system first establishes a supported finding, then checks the intervention contract, then measures downstream consequences, and finally reports a task-relative readiness decision. Each step has a distinct failure mode. A perfect missingness detector need not select beneficial imputation; a verified domain mask need not recover an original value; an acceptable repaired score need not justify modifying a native table. Native controls and signed effects make these distinctions observable.

Multi-metric evaluation is particularly useful in this evidence. Native LR improves macro-F1 while degrading ranking quality, and C05 loses substantial ROC-AUC without exceeding the macro-F1 tolerance. A readiness contract must therefore name its metrics and structural conditions rather than infer readiness from a single quality score. The two-point tolerance is a declared policy choice, not a universal standard. A dataset can satisfy it relative to a weak reference and remain inadequate for a real application.

For tool-using assessors, correct completed outputs are only one operational outcome. Completion, protected execution, and resource usage must be assessed against the same assignment ledger. The agent study's main result is the observed interaction of these outcomes, including failure to recognize ready cases. It does not show that a coordinator or specialist role caused the errors, nor that a language model is required to implement the framework.

B. Scope and validity

The principal utility slice contains 27 LR comparisons on three reused holdouts, not a full crossed comparison of all methods and learners. Historical repair policies differ, so their effects cannot be pooled into one treatment estimate. The additional Adult examples were selected after outcomes and should be read as illustrations. Observed sign counts and three-seed SDs do not establish population harm rates or statistically significant task-wide effects.

Synthetic corruption supplies exact truth for constructed defects but does not cover all natural semantic errors. Adult complete-case selection changes the evaluated population. Diabetes uses discharge-time features and recorded readmission with incomplete follow-up; patient separation is not hospital- or time-external validation. Steel's chronological partition improves temporal discipline but leaves source-label semantics, latency, seasonality, and longer dependence unresolved. No untouched external cohort has been evaluated.

The six-case readiness study is prospectively specified at the corruption level on an already used holdout, and has only two cases per class. Its small size cannot support broad accuracy estimates. Same observations and repair permissions do not control compute, prompting, or orchestration. The live cohort uses one language model, visible semantic metadata, and no controlled role-removal experiment. Provider dependence, metadata recognition, and causal specialist contribution remain unmeasured. Readiness-score correlations and probability calibration are unavailable because consistent numeric readiness scores and class probabilities were not collected.

Conditional bootstrap intervals in the original archive omit training variation, repeated-holdout selection, and simultaneous coverage. Independent refits verify calculations and do not add populations or cases. These limits constrain inference even when repairs and their recorded effects are reproducible.

VII. Conclusion

We evaluate dataset quality through a connected chain of defect detection, constrained repair, predictive utility, and reference-relative readiness. In the complete saved missingness slice, perfect cell detection coexists with consistently negative Adult effects, positive Diabetes effects, and predominantly negative Steel effects. Native controls and the all-predictors-missing condition demonstrate why structural checks and multiple task metrics both matter. In the six-case agent study, failure-inclusive correctness and completion give a less favorable operational picture than completed-only scores. Together, the findings support joint evaluation of evidence, edits, utility, and completion within a declared contract, while keeping architectural and deployment claims within the measured scope.

Reproducibility

The companion sources retain the archived cohort summaries, readiness configuration and ledger, source hashes, descriptive analysis, and vector figures needed to regenerate the presented numerical summaries. The compact bundle does not contain the complete historical raw archive or all saved held-out predictions; reproducing all original fits and uncertainty intervals requires that fuller research archive. The source datasets are attributed to their UCI records [7]–[9]. This synthesis adds no fitted model, provider request, or experimental observation. A public repository or archival DOI has not been supplied.

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